(7) OPERATORS :-

- Types of operators :-
  - Symbols that tell compiler to perform some operation

Expression → Sum = a + b

operator

operands.

- Arithmatic Operator (Binary / unary)
- Relational operators
- Logical operators
- Bitwise Operators (Bit manupulation)
- Assignment operators

- Arithmatic operator

Binary (2 operands)	unary
+	++
-	--
*	
/	
%	

→ Binary Operators :- (Let A = 10, B = 5)

+ → A + B = 10 + 5 = 15

− → A − B = 10 − 5 = 5

* → A * B = 10 * 5 = 50

/ → A / B = 10 / 5 = 2

% (modulo) → A % B = 10 % 5 = 0

↳ It tells about remainder

- Unary Operator :-

$a = a + 1$ ++ a++ or ++a = increment operator

$a = a - 1$ -- a-- or --a = Decrement operator

Case I

Pre increment	Post increment
++ a	a ++
1. Value change	1. Value use
2. Value use	2. Value change

Case II

Pre Decrement	Post Decrement
--a	a --
1. Value change	1. Value use
2. Value use	2. Value change

Code :-

```
import. java.util.*;
public class vishal {
    public static void main (String args[]) {
        int a = 10;
        int b = ++a;
        S.O.P ln (a);
        S.O.P ln (b);
    }
}
```

```
int a = 10
int b = a++
```

Output :-

11
10

Output :-

11
11

Relational operator.

== Equal $\overset{(10)}{A} == \overset{(10)}{B}$ → true $\overset{(10)}{A} == \overset{(5)}{B}$ → false

!= Not equal (inequality) $\underset{(10)}{A} != \underset{(5)}{B}$ → true $\underset{(10)}{A} != \underset{(10)}{B}$ → false

> Greater than $\underset{(10)}{A} > \underset{(5)}{B}$ → true $\underset{(10)}{A} > \underset{(15)}{B}$ → false

< less than $\underset{(5)}{A} < \underset{(10)}{B}$ → true $\underset{(10)}{A} < \underset{(5)}{B}$ → false

>= Greater than equal to

<= less than equal to

Logical operator:-

&& (Logical AND) → True
(S++1) && (S++2)
T T

A	Statement I	Stat II
T	T	T
T	T	F
F	F	T
F	F	F

|| (Logical OR) → True
(S++1) || (S++2)
T F

A	St. I	St II
T	T	T
T	T	F
T	F	T
F	F	F

! (Logical NOT) (Not operator)
true → false !(S++1)
false → true

A	St
F	T
T	F

Assignment operator:-

= A = B

+= } A = A+10 B = B-5
-= } A += 10 B -= 5

*= } A = A * 10 B = B 7.5
/= } A *= 10 B 7. = 5